

Unit 1: Rates of Change and Tangent Lines — Full Solutions

Warm-up

1. $f(x) = x^2$, from $x = 1$ to $x = 4$.

Compute the two values: $f(4) = 16$ and $f(1) = 1$.

$$\frac{f(4) - f(1)}{4 - 1} = \frac{16 - 1}{3} = \frac{15}{3} = 5$$

Average rate of change = 5.

2. $f(x) = 3x + 2$, from $x = -2$ to $x = 5$.

$$f(5) = 3(5) + 2 = 17. \quad f(-2) = 3(-2) + 2 = -4.$$

$$\frac{17 - (-4)}{5 - (-2)} = \frac{21}{7} = 3$$

Average rate of change = 3. Notice this is exactly the slope 3 from $f(x) = 3x + 2$ — for a straight line, the average rate of change between any two points is always the same, and it's just the line's slope.

3. $f(x) = x^2 - 2x$ on $[0, 3]$.

$$f(3) = 9 - 6 = 3. \quad f(0) = 0 - 0 = 0.$$

$$\frac{3 - 0}{3 - 0} = 1$$

Average rate of change = 1.

4. $f(x) = \sqrt{x}$, from $x = 1$ to $x = 9$.

$$f(9) = 3. \quad f(1) = 1.$$

$$\frac{3 - 1}{9 - 1} = \frac{2}{8} = \frac{1}{4}$$

Average rate of change = $\frac{1}{4}$.

5. Points $(2, 5)$ and $(6, 13)$.

$$\text{Slope} = \frac{\text{change in } y}{\text{change in } x} = \frac{13 - 5}{6 - 2} = \frac{8}{4} = 2.$$

Slope of secant line = 2.

6. $f(x) = x^3$, from $x = 1$ to $x = 2$.

$$f(2) = 8. \quad f(1) = 1.$$

$$\frac{8 - 1}{2 - 1} = 7$$

Slope of secant line = 7.

7. $f(x) = \frac{1}{x}$ on $[1, 2]$.

$$f(2) = \frac{1}{2}. \quad f(1) = 1.$$

$$\frac{\frac{1}{2} - 1}{2 - 1} = \frac{-\frac{1}{2}}{1} = -\frac{1}{2}$$

Average rate of change = $-\frac{1}{2}$.

Standard

8. $f(x) = x^2$, tangent slope at $x = 3$.

Step 1 — write $f(3+h)$: $(3+h)^2 = 9 + 6h + h^2$.

Step 2 — subtract $f(3) = 9$: $9 + 6h + h^2 - 9 = 6h + h^2$.

Step 3 — divide by h : $\frac{6h + h^2}{h} = 6 + h$ (we can cancel one h from every term, since $h \neq 0$ while we're doing this step).

Step 4 — let $h \rightarrow 0$: $6 + 0 = 6$.

Tangent slope at $x = 3$ is 6.

9. $f(x) = x^2 + 1$, tangent slope at $(1, 2)$.

$$f(1+h) = (1+h)^2 + 1 = 1 + 2h + h^2 + 1 = 2 + 2h + h^2.$$

Subtract $f(1) = 2$: $2 + 2h + h^2 - 2 = 2h + h^2$.

Divide by h : $2 + h$.

Let $h \rightarrow 0$: 2.

Tangent slope at $(1, 2)$ is 2.

10. $f(x) = 2x^2 - 3x$, tangent slope at $x = 2$.

$$f(2+h) = 2(2+h)^2 - 3(2+h).$$

Expand $(2+h)^2 = 4 + 4h + h^2$, so $2(2+h)^2 = 8 + 8h + 2h^2$.

Expand $-3(2+h) = -6 - 3h$.

Add them: $8 + 8h + 2h^2 - 6 - 3h = 2 + 5h + 2h^2$.

$$f(2) = 2(4) - 3(2) = 8 - 6 = 2.$$

Subtract: $2 + 5h + 2h^2 - 2 = 5h + 2h^2$.

Divide by h : $5 + 2h$.

Let $h \rightarrow 0$: 5.

Tangent slope at $x = 2$ is 5.

11. $f(x) = x^2$, tangent line at $x = -1$.

First find the slope: $f(-1) = 1$.

$$f(-1+h) = (-1+h)^2 = 1 - 2h + h^2.$$

$$\text{Subtract } f(-1) = 1: 1 - 2h + h^2 - 1 = -2h + h^2.$$

$$\text{Divide by } h: -2 + h.$$

Let $h \rightarrow 0$: -2 . So the slope is -2 .

Now use the point-slope formula with point $(-1, 1)$ and slope -2 :

$$y - 1 = -2(x - (-1)) = -2(x + 1)$$

$$y - 1 = -2x - 2$$

$$y = -2x - 1$$

Tangent line: $y = -2x - 1$.

12. $f(x) = -2x^2$, tangent slope and tangent line at $x = 1$.

$$f(1) = -2(1)^2 = -2.$$

$$f(1+h) = -2(1+h)^2 = -2(1+2h+h^2) = -2-4h-2h^2.$$

$$\text{Subtract } f(1) = -2: -2-4h-2h^2 - (-2) = -4h-2h^2.$$

$$\text{Divide by } h: -4-2h.$$

Let $h \rightarrow 0$: -4 . So the slope is -4 .

Tangent line through $(1, -2)$ with slope -4 :

$$y - (-2) = -4(x - 1)$$

$$y + 2 = -4x + 4$$

$$y = -4x + 2$$

Tangent slope $= -4$; tangent line: $y = -4x + 2$.

13. $f(x) = x^3$, tangent slope at $x = 1$.

$$f(1) = 1.$$

$$f(1+h) = (1+h)^3 = 1 + 3h + 3h^2 + h^3 \text{ (this comes from the pattern } (a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 \text{).}$$

Subtract $f(1) = 1$: $1 + 3h + 3h^2 + h^3 - 1 = 3h + 3h^2 + h^3$.

Divide by h : $3 + 3h + h^2$.

Let $h \rightarrow 0$: 3.

Tangent slope at $x = 1$ is 3.

Challenge

14. $f(x) = x^2 + 1$, tangent and normal line at $(1, 2)$.

From Problem 9, we already know the tangent slope at this point is 2.

Tangent line: $y - 2 = 2(x - 1) \Rightarrow y = 2x$.

Normal line: the normal slope is the negative reciprocal of 2, which is $-\frac{1}{2}$.

$$y - 2 = -\frac{1}{2}(x - 1)$$

$$y = -\frac{1}{2}x + \frac{1}{2} + 2 = -\frac{1}{2}x + \frac{5}{2}$$

Tangent line: $y = 2x$. Normal line: $y = -\frac{1}{2}x + \frac{5}{2}$.

15. $f(x) = x^2 - 4x + 3$, find where the tangent is horizontal.

A horizontal line has slope 0, so we need to find where the tangent slope equals 0. Let's find a general formula for the slope at any point $x = a$, then set it to 0.

$$f(a) = a^2 - 4a + 3.$$

$$f(a + h) = (a + h)^2 - 4(a + h) + 3 = a^2 + 2ah + h^2 - 4a - 4h + 3.$$

$$\text{Subtract } f(a): (a^2 + 2ah + h^2 - 4a - 4h + 3) - (a^2 - 4a + 3) = 2ah + h^2 - 4h.$$

Divide by h : $2a + h - 4$.

Let $h \rightarrow 0$: slope $= 2a - 4$.

Set the slope to 0: $2a - 4 = 0 \Rightarrow a = 2$.

Find the y -value there: $f(2) = 4 - 8 + 3 = -1$.

The tangent line is horizontal at the point $(2, -1)$.

16. $f(x) = x^2$, find where the tangent slope is 6.

Using the same style as Problems 8 and 11, the general slope formula for $f(x) = x^2$ at any point $x = a$ works out to $2a$ (you can check: at $a = 3$ we got slope 6 matching $2(3) = 6$ from Problem 8, and at $a = -1$ we got slope -2 matching $2(-1) = -2$ from Problem 11).

Set $2a = 6 \Rightarrow a = 3$.

Find y : $f(3) = 9$.

The tangent slope equals 6 at the point $(3, 9)$.

17. $f(x) = \frac{1}{x}$, tangent slope at $x = 2$.

$$\frac{f(2+h) - f(2)}{h} = \frac{\frac{1}{2+h} - \frac{1}{2}}{h}$$

To subtract the two fractions in the numerator, put them over a common denominator $2(2+h)$:

$$\frac{1}{2+h} - \frac{1}{2} = \frac{2 - (2+h)}{2(2+h)} = \frac{-h}{2(2+h)}$$

Now divide this whole thing by h (which means multiplying by $\frac{1}{h}$):

$$\frac{-h}{2(2+h)} \cdot \frac{1}{h} = \frac{-1}{2(2+h)}$$

Now let $h \rightarrow 0$:

$$\frac{-1}{2(2+0)} = \frac{-1}{4}$$

Tangent slope at $x = 2$ is $-\frac{1}{4}$.

18. $f(x) = \sqrt{x}$, tangent slope at $x = 4$.

$$\frac{f(4+h) - f(4)}{h} = \frac{\sqrt{4+h} - \sqrt{4}}{h} = \frac{\sqrt{4+h} - 2}{h}$$

Here, plugging in $h = 0$ directly gives $\frac{0}{0}$, and there's no obvious factor of h to cancel — so we use a special trick: multiply top and bottom by the conjugate $\sqrt{4+h} + 2$. This works because $(\sqrt{4+h} - 2)(\sqrt{4+h} + 2) = (4+h) - 4 = h$, which clears out the square root.

$$\frac{\sqrt{4+h} - 2}{h} \cdot \frac{\sqrt{4+h} + 2}{\sqrt{4+h} + 2} = \frac{(4+h) - 4}{h(\sqrt{4+h} + 2)} = \frac{h}{h(\sqrt{4+h} + 2)}$$

Cancel the h :

$$\frac{1}{\sqrt{4+h} + 2}$$

Now let $h \rightarrow 0$:

$$\frac{1}{\sqrt{4+0} + 2} = \frac{1}{2+2} = \frac{1}{4}$$

Tangent slope at $x = 4$ is $\frac{1}{4}$.

Applied

19. $h(t) = 20t - 5t^2$, average velocity from $t = 1$ to $t = 3$.

$$h(3) = 20(3) - 5(9) = 60 - 45 = 15.$$

$$h(1) = 20(1) - 5(1) = 20 - 5 = 15.$$

$$\frac{h(3) - h(1)}{3 - 1} = \frac{15 - 15}{2} = 0$$

Average velocity = 0 m/s. This makes sense: the ball is at the same height (15 m) at both times — it went up and came back down to that same height, so its net change in position over that interval is zero, even though it was clearly moving the whole time.

20. Instantaneous velocity at $t = 1$ for $h(t) = 20t - 5t^2$.

We'll use k as our tiny step (to avoid confusing it with the function's name h).

$$h(1) = 15 \text{ (from Problem 19).}$$

$$\begin{aligned} h(1+k) &= 20(1+k) - 5(1+k)^2 = 20 + 20k - 5(1 + 2k + k^2) = 20 + 20k - 5 - 10k - 5k^2 \\ &= 15 + 10k - 5k^2 \end{aligned}$$

Subtract $h(1) = 15$: $10k - 5k^2$.

Divide by k : $10 - 5k$.

Let $k \rightarrow 0$: 10.

Instantaneous velocity at $t = 1$ is 10 m/s.

21. $s(t) = t^2 + 2t$, average speed on $[0, 2]$ and instantaneous speed at $t = 2$.

Average speed: $s(2) = 4 + 4 = 8$. $s(0) = 0$.

$$\frac{8 - 0}{2 - 0} = 4$$

Average speed = 4 mph.

Instantaneous speed at $t = 2$:

$$s(2+k) = (2+k)^2 + 2(2+k) = 4 + 4k + k^2 + 4 + 2k = 8 + 6k + k^2$$

Subtract $s(2) = 8$: $6k + k^2$.

Divide by k : $6 + k$.

Let $k \rightarrow 0$: 6.

Instantaneous speed at $t = 2$ is 6 mph.

22. $T(x) = -0.1x^2 + 2x + 50$. Average rate of change from $x = 2$ to $x = 8$, and instantaneous rate at $x = 5$.

Average rate of change:

$$T(8) = -0.1(64) + 2(8) + 50 = -6.4 + 16 + 50 = 59.6.$$

$$T(2) = -0.1(4) + 2(2) + 50 = -0.4 + 4 + 50 = 53.6.$$

$$\frac{59.6 - 53.6}{8 - 2} = \frac{6}{6} = 1$$

Average rate of change = 1°F per hour.

Instantaneous rate at $x = 5$:

$$T(5+k) = -0.1(5+k)^2 + 2(5+k) + 50$$

Expand $(5+k)^2 = 25 + 10k + k^2$, so $-0.1(5+k)^2 = -2.5 - k - 0.1k^2$.

Expand $2(5+k) = 10 + 2k$.

Add everything: $-2.5 - k - 0.1k^2 + 10 + 2k + 50 = 57.5 + k - 0.1k^2$.

$$T(5) = -0.1(25) + 10 + 50 = -2.5 + 10 + 50 = 57.5.$$

Subtract: $k - 0.1k^2$.

Divide by k : $1 - 0.1k$.

Let $k \rightarrow 0$: 1.

Instantaneous rate of change at hour 5 is 1°F per hour.